

Topos-Theoretic Bridges: Transferring Results Across Case-Theoretic Frameworks

The Inter-Theoretic Translation Problem

The preceding sections constructed four formally distinct perspectives on case: typological, type-logical, distributional, and enriched. A central methodological question arises: *when can structural results proved in one framework be carried over to another without starting from scratch?* Caramello's [-@caramello2016bridges] topos-theoretic bridge technique answers this for properties that are **invariants of a shared classifying topos**, once Morita equivalence or a suitable bridge is in hand.

Classifying Toposes: The Logical Shape of a Theory

A Topos Is a Universe With Internal Logic

A **topos** is a category possessing the structural richness of a generalized “universe of sets”: products, exponentials, and a subobject classifier (a “truth-value object”) that supports internal first-order reasoning. The most familiar example is the category **Set** of ordinary sets; other important instances include presheaf categories $[\mathcal{C}^{\text{op}}, \mathbf{Set}]$ and sheaf categories over topological spaces. Intuitively, a topos provides a *self-contained logical universe* within which mathematical reasoning can proceed—and different toposes encode different logical constraints.

Morita Equivalence: Invariant Transfer Across Typologies

Every *geometric theory* $\mathbb{T}$ (axiomatized by sequents with finite conjunctions, arbitrary disjunctions, and existential quantification) generates a unique **classifying topos** $\mathcal{E}_{\mathbb{T}}$: a canonical topos whose models correspond one-to-one with geometric functors out of $\mathcal{E}_{\mathbb{T}}$. The classifying topos encodes the theory’s “logical shape” independently of any particular model.

Caramello’s insight [–@caramello2016bridges; –@caramello2021five]

A Chain of Morita Equivalences Connects Case Theories

Formalizing Case Theories as Geometric Theories

We formalize each case-theoretic framework as a geometric theory:

1. **Typological case theory** $\mathbb{T}_{\text{typ}}$: Objects are case roles, morphisms are grammatical relations, axioms specify alignment constraints (e.g., “S = A” in accusative alignment).
2. **Type-logical case theory** $\mathbb{T}_{\text{log}}$: Objects are syntactic types, morphisms are Lambek calculus derivations, axioms specify well-formedness conditions (pregroup contractions).
3. **Distributional case theory** $\mathbb{T}_{\text{dist}}$: Objects are vector spaces, morphisms are linear maps, axioms specify the composition law for distributional meaning (DisCoCat).
4. **Enriched case theory** $\mathbb{T}_{\text{enr}}$: Objects are expressions, morphisms carry $[0, 1]$ -valued weights, axioms specify the identity and composition inequalities.

The Bridge Theorem: Morita Equivalence Chain

The **research programme** we pursue is that these four perspectives admit a topos-theoretic alignment: formally distinct case theories should be related by bridge toposes and, where

Phillips: LoT Properties Arise as Universal Topos Constructions

Phillips [-@phillips2024lot] provides a striking application of topos-theoretic methods to cognitive science. He shows that the **Language of Thought** (LoT) hypothesis—the claim that cognition operates over structured, combinatorial representations with language-like properties—can be formalized categorically, and that the resulting structure is *universal* in the topos-theoretic sense.

Specifically, Phillips demonstrates that:

1. LoT properties (discrete constituents, role-filler independence, systematicity) arise as **universal constructions** in a topos—categorical products, fiber bundles, and presheaves.
2. Every topos supports an internal first-order logic, explaining how LoT-like logical capacities can emerge in systems (biological or artificial) whose internal architecture forms a topos.
3. The “shape” of cognitive representations is fundamentally

Caramello's Learning Algorithm

Caramello [-@caramello2023syntactic] extends the bridge technique to a learning theory: she shows that classifying toposes can be used to *learn* the theory of a mathematical structure from finite data (a finite set of models). The learning algorithm constructs a classifying topos from the observed data and then extracts the axioms of the underlying theory.

Applied to case systems, this suggests a principled approach to *grammatical induction*: given a corpus annotated with case labels, one could construct the classifying topos of the implicit case theory and read off its axioms—recovering the alignment type, the morphism structure, and the enriched weights from data alone. The procedure operates in four phases:

1. **Extraction**: Parse a Universal Dependencies treebank for a target language, collecting all case-labeled dependency arcs. Each arc (r_1, rel, r_2) instantiates a morphism $r_1 \rightarrow r_2$ in the implicit case category.
2. **Saturation**: Close the extracted morphism set under

Morita Equivalence Diagrams Are Themselves Free-Ride Inferences

The bridge technique has a natural diagrammatic interpretation. Morita equivalence between theories is witnessed by *functorial translations*—diagrams in the 2-category of toposes that commute up to natural isomorphism. These diagrams serve the same cognitive function as the commutative diagrams of **??**: they make the transfer of structure visible, allowing a researcher to verify at a glance that a result proved in one framework genuinely applies in another.

Manders [-@manders2008euclidean] observed that even in classical mathematics, diagrams serve not merely as illustrations but as *inferential instruments* whose spatial properties encode proof-relevant information. The topos-theoretic bridge diagrams extend this observation to the meta-theoretical level: the commutative diagram expressing Morita equivalence is itself a “free ride” inference, automatically transferring any topos-invariant property from one theory to another without requiring a

Python Implementation: `build_typological_theory()`, `check_morita_equivalence()`, `bridge_transfer()`—All Working

The topos-theoretic narrative above is paired with a topos Python module that implements **finite geometric theories** extracted from `CaseCategory`, **invariant-profile comparison** (`check_morita_equivalence`), and **guarded bridge transfer** when profiles match—not a full classifying-topos construction inside the runtime.

Geometric Theories from Case Categories

The `build_typological_theory()` function constructs a geometric theory $\mathbb{T}_{\text{typ}}$ from a `CaseCategory` by extracting:

- ▶ **Sorts**: the case roles (objects of the category)
- ▶ **Function symbols**: the morphisms with their source/target pairs
- ▶ **Axioms**: identity morphism existence, composition closure, and alignment constraints

For the standard 8-case category, this yields a theory with 8 sorts